

Xiyan Zhou

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EDUCATION

University of Michigan, Ann Arbor

Ann Arbor, MI

B.S. Computer Science & B.S. Mathematics (Double Major)

GPA: 3.8 / 4.0

Coursework: Measure Theory, Convex Optimization, Bayesian Machine Learning, Numerical Linear Algebra, Real Analysis, Probability Theory, Stochastic Processes, Machine Learning, Operating Systems, Data Structures & Algorithms, Combinatorics, Discrete Mathematics

SKILLS

Programming Languages: Python, C++, C, Java, SQL, MATLAB, JavaScript

Frameworks & Tools: JAX, PyTorch, NumPy, scikit-learn, L^AT_EX, Git, Bash, Linux

ML & Math: Reinforcement Learning, Bayesian Methods, Convex Optimization, CNNs, Statistical Modeling

Languages: Mandarin (native), English (proficient)

RESEARCH

Motion Planning via Contrastive Reinforcement Learning

Research Assistant — advised by Kellen Kanarios (Lei Ying Lab)

Jan 2026 – Present

- Studying goal-conditioned contrastive RL with a Gumbel-Softmax actor for discrete action spaces, extending the JaxGCRL framework by replacing continuous probability distributions with categorical logits.
- Investigating environments where picked-up objects deplete and respawn at random locations, turning the task into a within-episode distribution-shift problem; reproducing a failure mode of contrastive RL flagged in a recent ICML 2026 paper on procedurally generated, abstract-goal settings.
- Implementing double Q-learning, with a hierarchical extension planned; validating agent performance on Navix grid-world environments and targeting a workshop submission.

AI-Driven Human Digital Twin Study (IOE SURE Program)

Undergraduate Research Assistant — advised by Prof. Jingwen Hu

May 2026 – Present

- Developing the lower extremity muscle segmentation component of the HERMES human body model; identified that TotalSegmentator, the standard public CT segmentation tool, only segments thigh muscles into broad compartments rather than individual muscles.
- Determined that compartment-level segmentation is insufficient for this application, since individual muscles within a compartment differ in fiber direction (relevant to force analysis) and cannot be separately evaluated for cross-sectional area.
- Hand-labeling individual thigh muscles from CT data to build a training set, and preparing to fine-tune the nnU-Net architecture underlying TotalSegmentator so it can automatically segment individual muscles going forward, potentially aided by transfer learning.

PROJECTS

Replicating and Extending SimGRL: Robustness in Visual RL under Dynamic Scene Perturbations

- Replicated the SimGRL visual reinforcement learning framework using PyTorch and DMControl to validate robustness against dynamic scene perturbations.
- Designed and implemented four novel data augmentation extensions (Irregular-SRO, Multi-Source SRO, Hybrid-SRO, Mask-Guided) to mitigate observational overfitting and enhance agent generalization.
- Conducted ablation studies on the Great Lakes HPC cluster, analyzing trade-offs between augmentation complexity and sample efficiency under strict computational constraints.
- Demonstrated that the Hybrid-SRO extension significantly reduced performance variance compared to the baseline, establishing a more stable training protocol for high-dimensional visual control tasks.

Predicting ICU Mortality

- Built a logistic regression model to predict 30-day ICU mortality using engineered features such as instability scores, crossover counts, and time-windowed statistics.
- Applied KNN imputation, feature normalization, and best-K feature selection to reduce model complexity; tuned regularization, class weights, and decision thresholds to balance AUROC and F1-score for clinical relevance.

AWARDS & TEACHING

- James B. Angell Scholar** — University of Michigan (March 2025); awarded to students with a perfect 4.0 term GPA
- University Honors** — University of Michigan (×4: Fall 2023, Winter 2024, Fall 2024, Winter 2025)
- Grader** — MATH 571 (Graduate Numerical Linear Algebra) and MATH 371; **Peer Tutor**, Math Learning Center (4 semesters)